Time period	Technological milestone/development phase	Key innovations introduced	Impact/significance
		integrated AI decision-making, autonomous swarm control, and stealth optimization.	
2022–2024	Emergence of quantum communication and edge computing	Incorporation of quantum key distribution (QKD), neural network processors, and digital twin-based predictive maintenance.	Enhanced UAV security, reliability, and adaptability through real-time, decentralized computation.
Mid-2020s	Evolution of smart materials and energy systems	Application of self-healing composites, morphing wings, and hybrid-electric propulsion systems.	Increased endurance, reduced radar signature, and improved flight survivability under harsh conditions.
2025–2027	Expansion into civilian and multi-domain operations	Integration into urban air mobility (UAM),	Diversified UAV applications beyond defense, promoting